

Assignment

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DIV : SE1

Batch : S15

Topic- Sales Dataset

Problem Statement:

A dataset representing sales transactions is provided. It contains Order ID, Product, Category, Region, Sales Rep, Month, Year, Quantity, Unit Price, Discount%, Sales Amount, Profit, Returns, and Customer Rating. The objective is to perform data analysis and manipulation using Pandas and NumPy in Python.

Dataset Creation & Preview

```
import numpy as np
```

```
import pandas as pd
```

```
np.random.seed(0)
```

```
products = ["Laptop","Smartphone","Tablet","Headphones","Smartwatch",  
            "Camera","Printer","Monitor","Keyboard","Mouse"]
```

```
regions = ["North","South","East","West"]
```

```
n = 300
```

```
qty_arr = np.random.randint(1, 50, n)
```

```
discount_arr = np.round(np.random.uniform(0, 0.3, n), 2)
```

```
sales_arr = np.round(price_arr * qty_arr * (1 - discount_arr), 2)
```

```
# Inject NaN into Sales_Amount and Profit
```

```
sales_arr[np.random.choice(n, 8, replace=False)] = np.nan
```

```
df = pd.DataFrame({  
    "Order_ID"      : [f"ORD{1000+i}" for i in range(n)],  
    "Product"       : prod_arr,  
    "Category"      : category_arr,  
    "Region"        : np.random.choice(regions, n),  
    "Sales_Amount"  : sales_arr,  
    "Profit"        : profit_arr,  
    "Customer_Rating": np.round(np.random.uniform(1,5,n),1),  
})
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code> python -u "c:\Users\ishan\OneDrive\Desktop\C tutorials\practice code\assignment.py"
===== DATASET PREVIEW =====
  Order_ID  Product  Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
0  ORD1000   Printer  Electronics   East   Riya   Aug 2023     41     12061     0.11      0      1.1     440105.89    88673.59
1  ORD1001  Headphones  Accessories  South   Sara   Feb 2023     17     42399     0.07      0      1.3     670328.19   184236.02
2  ORD1002   Monitor  Electronics   West   Riya   Sep 2022     40      1025     0.23      1      4.6     31570.00     6273.25
3  ORD1003  Smartwatch  Electronics  North   Riya   Jan 2022     31     23361     0.08      1      1.6     666255.72   160202.36
4  ORD1004   Printer  Accessories   West   Riya   Aug 2022     34     49653     0.25      0      3.1    1266151.50   378026.46
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

01 Find total number of records and features, and display all column names.

```
r, c = df.shape  
print(f"Column names: {list(df.columns)}")
```

```
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3  ORD1003  Smartwatch  Electronics  North   Riya   Jan 2022     31     23361     0.08      1      1.6     666255.72   160202.36
4  ORD1004   Printer  Accessories   West   Riya   Aug 2022     34     49653     0.25      0      3.1    1266151.50   378026.46
Q1
300 14 ['Order_ID', 'Product', 'Category', 'Region', 'Sales_Rep', 'Month', 'Year', 'Quantity', 'Unit_Price', 'Discount', 'Returns', 'Customer_Rating', 'Sales_Amount', 'Profit']
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

02

Calculate total, average, minimum, and maximum Sales Amount.

```
sa = df["Sales_Amount"].dropna().values  
print(f"Max Sales : ₹{np.max(sa):>15,.2f}")
```

```

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Order_ID Product Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
0 ORD1000 Printer Electronics East Riya Aug 2023 41 12061 0.11 0 1.1 440105.89 88673.59
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4 ORD1004 Printer Accessories West Riya Aug 2022 34 49653 0.25 0 3.1 1266151.50 378026.46

Q2
160179949.1 548561.4695205479 575.0 1987461.56
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

03

Identify and count missing values in each column of the dataset.

```
mv = df.isnull().sum()
```

```
print(f"\nTotal missing cells: {mv.sum()}")
```

```

PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code> python -u "c:\Users\ishan\OneDrive\Desktop\C tutorials\practice code\assignment.py"
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3 ORD1003 Smartwatch Electronics North Riya Jan 2022 31 23361 0.08 1 1.6 666255.72 160202.36
4 ORD1004 Printer Accessories West Riya Aug 2022 34 49653 0.25 0 3.1 1266151.50 378026.46

Q3
Order_ID      0
Product       0
Category      0
Region        0
Sales_Rep     0
Month         0
Year          0
Quantity      0
Unit_Price    0
Discount      0
Returns       0
Customer_Rating 0
Sales_Amount  8
Profit        8
dtype: int64

```

04

Find the product with the highest total sales amount.

```
ps = df.groupby("Product")["Sales_Amount"].sum().sort_values(ascending=False)
```

```
print(f"\nBest Product: {ps.idxmax()} → ₹{ps.max():.2f}")
```

```

PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code> python -u "c:\Users\ishan\OneDrive\Desktop\C tutorials\practice code\assignment.py"
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4 ORD1004 Printer Accessories West Riya Aug 2022 34 49653 0.25 0 3.1 1266151.50 378026.46

Q4
Printer 23478349.169999998
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

05 Compute total profit per region and identify the most profitable region.

```
rp = df.groupby("Region")["Profit"].sum().sort_values(ascending=False)
```

```
print(f"\nMost Profitable Region: {rp.idxmax()} → ₹{rp.max():.2f}")
```

```

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4  ORD1004  Printer  Accessories West      Riya  Aug 2022      34      49653      0.25      0          3.1      1266151.50  378026.46

Q5
North 11895082.59
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

06 Calculate standard deviation and variance of Quantity sold.

```
q = df["Quantity"].values
```

```
print(f"Variance (Pandas): {df['Quantity'].var():.4f}")
```

```

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4  ORD1004  Printer  Accessories West      Riya  Aug 2022      34      49653      0.25      0          3.1      1266151.50  378026.46

Q6
13.25644497342079 175.73333333333332
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

07 Add a new column 'Profit_Margin_%' = (Profit / Sales_Amount) × 100.

```
df["Profit_Margin_%"] = (df["Profit"] / df["Sales_Amount"] * 100).round(2)
```

```
print(f"\nAvg Profit Margin: {df['Profit_Margin_%'].mean():.2f}%")
```

```

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  Order_ID  Product  Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
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Q7
  Profit_Margin_%
0      20.148240
1      27.484451
2      19.870922
3      24.045176
4      29.856337
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

08 Classify sales orders as 'High', 'Medium', or 'Low' using pd.cut().

```
df["Sales_Category"] = pd.cut(df["Sales_Amount"],
```

```
    bins=[0, 500000, 1500000, df["Sales_Amount"].max()+1],
```

```
    labels=["Low", "Medium", "High"])
```

```
print(df[["Product", "Sales_Amount", "Sales_Category"]].dropna().head(8))
```

```

===== DATASET PREVIEW =====
  Order_ID  Product  Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
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Q8
Sales_Category
Low      164
Medium   117
High      11
Name: count, dtype: int64
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

09 Find the top 3 Sales Representatives by total sales amount.

```
sr = df.groupby("Sales_Rep")["Sales_Amount"].sum().sort_values(ascending=False)

print(f"\nTop 3 Reps: {list(sr.head(3).index)}")
```

```
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Q9
Sales_Rep
Amit      37109720.35
John      34510401.85
Sara       30019555.41
Name: Sales_Amount, dtype: float64
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

10 Compute 25th, 50th, and 75th percentile of Sales Amount using NumPy.

```
sv = df["Sales_Amount"].dropna().values

p25, p50, p75 = np.percentile(sv, [25, 50, 75])

print(f"IQR : ₹{p75-p25:>12.2f}")
```

```
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Q10
[190120.035 426158.06 828873.63 ]
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

11 Find correlation between Sales Amount, Profit, and Quantity sold.

```
cc = ["Sales_Amount", "Profit", "Quantity"]

print(df[cc].corr().round(3))

a = df[cc].dropna().values

print(f"NumPy corr(Sales, Qty) : {np.corrcoef(a[:,0], a[:,2])[0,1]:.3f}")
```

```
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Q11
Sales_Amount  Profit  Quantity
Sales_Amount  1.000000  0.909723  0.642461
Profit         0.909723  1.000000  0.548525
Quantity       0.642461  0.548525  1.000000
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

12 Normalize the Sales Amount column to a 0–1 scale using NumPy min-max.

```
mask = df["Sales_Amount"].notna()

sv2 = df.loc[mask, "Sales_Amount"].values
```

```
df.loc[mask, "Sales_Norm"] = ((sv2 - sv2.min()) / (sv2.max() - sv2.min())).round(4)
```

```
print(df[["Product", "Sales_Amount", "Sales_Norm"]].dropna().head(10))
```

```
===== DATASET PREVIEW =====
Order_ID Product Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
0 ORD1000 Printer Electronics East Riya Aug 2023 41 12061 0.11 0 1.1 440105.89 88673.59
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Q12
Sales_Norm
0 0.221216
1 0.337087
2 0.015600
3 0.335037
4 0.636965
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

13 Find month-wise total sales and identify the best performing month.

```
month_order = ["Jan", "Feb", "Mar", "Apr", "May", "Jun",
               "Jul", "Aug", "Sep", "Oct", "Nov", "Dec"]
```

```
ms = df.groupby("Month")["Sales_Amount"].sum().reindex(month_order)
```

```
print(f"\nBest Month: {ms.idxmax()} → ₹{ms.max():.2f}")
```

```
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Q13
Month
Apr 11869583.00
Aug 12162904.57
Dec 13167380.25
Feb 17625887.04
Jan 15454542.44
Jul 16812449.71
Jun 13200474.00
Mar 12044480.18
May 15657684.52
Nov 10420146.22
Oct 11988902.41
Sep 9775514.76
Name: Sales_Amount, dtype: float64
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

14 Detect outliers in Profit column using the IQR method.

```
q1 = df["Profit"].quantile(0.25)
```

```
q3 = df["Profit"].quantile(0.75)
```

```
iqr = q3 - q1
```

```
lo, hi = q1 - 1.5*iqr, q3 + 1.5*iqr
```

```
out = df[(df["Profit"] < lo) | (df["Profit"] > hi)]
```

```
print(out[["Product", "Profit"]].head(5))
```

```
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Q14
10
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

15

Calculate year-wise total sales, average sales, and order count.

```
ys = df.groupby("Year")["Sales_Amount"].agg(["sum", "mean", "count"]).round(2)
ys.columns = ["Total_Sales", "Avg_Sales", "Orders"]
print(ys.to_string())
```

```
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Q15
      sum      mean  count
Year
2022  63417869.23  592690.366636  107
2023  49372912.14  548587.912667   90
2024  47389167.73  498833.344526   95
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

16

Find the percentage of returned orders vs non-returned orders.

```
ret_pct = df["Returns"].value_counts(normalize=True) * 100
print(f"Returned : {ret_pct[1]:.2f}%")
print(f"Not Returned: {ret_pct[0]:.2f}%")
arr = df["Returns"].values
print(f"NumPy return count: {np.sum(arr)} / {len(arr)}")
```

```
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4  ORD1004  Printer  Accessories  West  Riya  Aug  2022  34  49653  0.25  0  3.1  1266151.50  378026.46

Q16
Returns
1    52.666667
0    47.333333
Name: proportion, dtype: float64
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```

17

Apply log transformation on Sales Amount to reduce right skewness.

```
sv3 = df["Sales_Amount"].dropna().values
df.loc[df["Sales_Amount"].notna(), "Log_Sales"] = np.log1p(sv3).round(4)
print(df[["Product", "Sales_Amount", "Log_Sales"]].dropna().head(8))
```

```

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4 ORD1004 Printer Accessories West Riya Aug 2022 34 49653 0.25 0 3.1 1266151.50 378026.46

Q17
Log_Sales
0 12.994773
1 13.415524
2 10.359994
3 13.409430
4 14.051493
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

18

Find average, min, and max Customer Rating per product category.

```
cr = df.groupby("Category")["Customer_Rating"].agg(["mean", "min", "max"]).round(2)
```

```
cr.columns = ["Avg_Rating", "Min_Rating", "Max_Rating"]
```

```
print(f"\nHighest Rated: {cr['Avg_Rating'].idxmax()} → {cr['Avg_Rating'].max():.2f}")
```

```

===== DATASET PREVIEW =====
Order_ID Product Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
0 ORD1000 Printer Electronics East Riya Aug 2023 41 12061 0.11 0 1.1 440105.89 88673.59
1 ORD1001 Headphones Accessories South Sara Feb 2023 17 42399 0.07 0 1.3 670328.19 184236.02
2 ORD1002 Monitor Electronics West Riya Sep 2022 40 1025 0.23 1 4.6 31570.00 6273.25
3 ORD1003 Smartwatch Electronics North Riya Jan 2022 31 23361 0.08 1 1.6 666255.72 160202.36
4 ORD1004 Printer Accessories West Riya Aug 2022 34 49653 0.25 0 3.1 1266151.50 378026.46

Q18
      mean  min  max
Category
Accessories 3.005031 1.0 5.0
Electronics 3.016312 1.1 4.9
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

19

Filter high-value orders (Sales > ₹10,00,000) and count by region.

```
hv = df[df["Sales_Amount"] > 1000000]
```

```
print(hv["Region"].value_counts().to_string())
```

```

===== DATASET PREVIEW =====
Order_ID Product Category Region Sales_Rep Month Year Quantity Unit_Price Discount Returns Customer_Rating Sales_Amount Profit
0 ORD1000 Printer Electronics East Riya Aug 2023 41 12061 0.11 0 1.1 440105.89 88673.59
1 ORD1001 Headphones Accessories South Sara Feb 2023 17 42399 0.07 0 1.3 670328.19 184236.02
2 ORD1002 Monitor Electronics West Riya Sep 2022 40 1025 0.23 1 4.6 31570.00 6273.25
3 ORD1003 Smartwatch Electronics North Riya Jan 2022 31 23361 0.08 1 1.6 666255.72 160202.36
4 ORD1004 Printer Accessories West Riya Aug 2022 34 49653 0.25 0 3.1 1266151.50 378026.46

Q19
Region
North 16
South 13
East 10
West 7
Name: count, dtype: int64
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>

```

20

Rank products within each category by total profit using Pandas rank().

```
cat_profit = df.groupby(["Category", "Product"])["Profit"].sum().reset_index()

cat_profit["Rank"] = cat_profit.groupby("Category")["Profit"].rank(
    ascending=False, method="dense")

cat_profit = cat_profit.sort_values(["Category", "Rank"])

print(cat_profit.to_string(index=False))
```

```
===== DATASET PREVIEW =====
Order_ID  Product  Category  Region  Sales_Rep  Month  Year  Quantity  Unit_Price  Discount  Returns  Customer_Rating  Sales_Amount  Profit
0  ORD1000  Printer  Electronics  East  Riya  Aug  2023  41  12061  0.11  0  1.1  440105.89  88673.59
1  ORD1001  Headphones  Accessories  South  Sara  Feb  2023  17  42399  0.07  0  1.3  670328.19  184236.02
2  ORD1002  Monitor  Electronics  West  Riya  Sep  2022  40  1025  0.23  1  4.6  31570.00  6273.25
3  ORD1003  Smartwatch  Electronics  North  Riya  Jan  2022  31  23361  0.08  1  1.6  666255.72  160202.36
4  ORD1004  Printer  Accessories  West  Riya  Aug  2022  34  49653  0.25  0  3.1  1266151.50  378026.46

Q20
   Category  Product  Profit  Rank
0  Accessories  Camera  357430.55  10.0
1  Accessories  Headphones  1889532.66  5.0
2  Accessories  Keyboard  1855586.99  6.0
3  Accessories  Laptop  2193169.87  4.0
4  Accessories  Monitor  2240878.61  3.0
5  Accessories  Mouse  1494931.75  8.0
6  Accessories  Printer  2308755.36  2.0
7  Accessories  Smartphone  1649884.32  7.0
8  Accessories  Smartwatch  899495.54  9.0
9  Accessories  Tablet  2779475.75  1.0
10 Electronics  Camera  1807245.37  3.0
11 Electronics  Headphones  1765205.58  4.0
12 Electronics  Keyboard  1221861.43  8.0
13 Electronics  Laptop  1494105.48  5.0
14 Electronics  Monitor  942444.29  9.0
15 Electronics  Mouse  732166.69  10.0
16 Electronics  Printer  2187797.59  1.0
17 Electronics  Smartphone  1260019.57  7.0
18 Electronics  Smartwatch  1928295.71  2.0
19 Electronics  Tablet  1365892.59  6.0
PS C:\Users\ishan\OneDrive\Desktop\C tutorials\practice code>
```